HILL OF TOWIE WIND FARM

AI-Based Performance Analysis and Power Prediction

Final Comprehensive Technical Report

EXECUTIVE SUMMARY & KEY RESULTS

- Facility: 21 Siemens SWT-2.3-VS-82 Turbines (48.3 MW Fleet Capacity), Moray, Scotland.
- Data Volume: 215 CSV files (9.30 GB) covering 16 continuous months (2025 Jan - 2026 Apr
- 2025 Annual Energy Yield: 96.96 GWh (Capacity Factor: 23.88%, Availability: 98.42%).
- Power Prediction Model: HistGradientBoostingRegressor (Selected Best Architecture).
- Unseen 2026 Holdout Test: R^2 = 0.9962, RMSE = 43.06 kW (1.87% Normalized RMSE), MAE
- Leakage Quarantine: Electrical current and operational flags strictly excluded from training
- Abnormal Behavior: 23 sustained underperformance events (>= 30 min, deficit <= -250 kW
- Alarm Correlation: 65.2% coincident with SCADA alarm logs; 34.8% isolated as silent aerod
- SCADA Replay Engine: Real-time streaming throughput of 675 steps/sec (1.48 ms latency).
- Decision Dashboard: Complete Dash + Plotly Desktop application with 7 interactive module

Metric Name	Empirical Value	Unit	Verification Status
Total Turbines	21	Units	VERIFIED
Farm Rated Capacity	48.3	MW	VERIFIED
2025 Annual Yield	96.96	GWh	VERIFIED
Capacity Factor	23.88	%	VERIFIED
Technical Availability	98.42	%	VERIFIED
Holdout Model R^2	0.9962	Ratio	VERIFIED
Holdout RMSE	43.06	kW	VERIFIED
Holdout NRMSE	1.87	%	VERIFIED
Sustained Anomalies	23	Events	VERIFIED
Replay Mean Latency	1.48	ms	VERIFIED

Key Project Analytical Outputs







